Tail-GAN

Learning to Simulate Tail Risk Scenarios

Chao Zhang
HKUST(GZ)

with Rama Cont (Oxford), Mihai Cucuringu (UCLA), Renyuan Xu (NYU)

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Motivation and Introduction

Score Functions for Risk Measures

Tail-GAN: Design and Theoretical Guarantee

Numerical Performance

TAiLgan
Learning to simulate tail Risk scenarios

Market simulator

What is a market simulator?

- Denote $\mathbb{P}_r$ as the target distribution of some price dynamics over a given time period. Given some samples $\{\rho_i\}_{i=1}^N$ from $\mathbb{P}_r$, the **goal** is to construct $\mathbb{P}_G$ such that

$$\mathbb{P}_G \text{ is "similar" to } \mathbb{P}_r \text{ under certain criterion}$$

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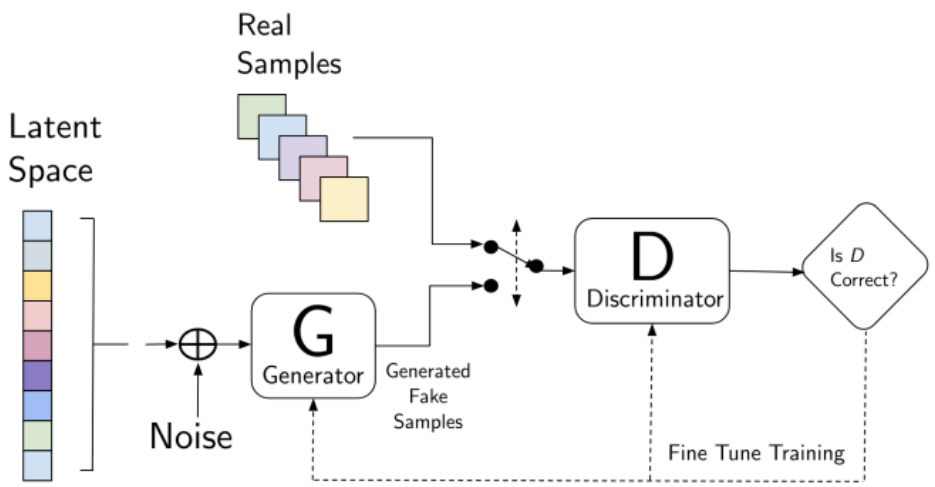
Why do we need a market simulator?

- Key ingredient of quantitative risk management systems.
- Main application: estimate the loss distributions for various trading strategies and portfolios.
- One focus is the calculation of **risk measures**, such as Value-at-Risk and Expected Shortfall.
- Other applications:
  - back-testing and stress testing; improve risk management; portfolio construction; forecasting
  - data protection

Classic simulation models in mathematical finance

- **Parametric approach**: time series models such as GARCH models, diffusion and jump processes, stochastic volatility models, and copulas
  - **Pros:** successful in low-dimensional applications involving a small number of homogeneous assets
  - **Cons:** difficult to model the joint dynamics of large or heterogeneous portfolios with assets from different classes $\Rightarrow$ lack of scalability + risk of model misspecification
- **Gaussian models** ("Black–Scholes models")
  - Default approach in most high-dimensional applications
  - **Cons:** mismatch with stylized facts observed from financial data (e.g., heavy tails, volatility clustering)

Generative Adversarial Networks (GAN)



Goodfellow et al. [2014]

- Robust network used for unsupervised machine learning to build a min-max game between two players: D (the discriminator) and G (the generator)
- Player G tries to fool player D by producing very "natural looking" data from random latent vector z
- Player D gets better at distinguishing between real and generated data

GAN in finance...

- GANs have been used for building 'market simulators in finance(e.g.Koshiyama et al. [2020], Wiese et al. [2020], Buehler et al. [2020], Yoon et al. [2019])by training them onraw returns.
- Training criterion is cross-entropy or some probability metrics, e.g., Wasserstein distance.
- Final result is a 'black-box' simulator that generates market scenarios.

Reference	Model category	Architecture	Loss Function	Data
Quant-GAN [Wiese et al., 2020]	GAN	TCN	KL divergence	Returns
[Buehler et al., 2020]	VAE	NN	l_2 norm	Signatures of paths
Time-GAN [Yoon et al., 2019]	GAN + Supervised Learning	RNN	Combination of log-likelihood and l_2 norm	Time Series
[Fu et al., 2019, Koshiyama et al., 2020]	CGAN	NN	KL divergence	Returns
SigCWGAN [Ni et al., 2020]	CGAN	Residual NN	l_2 norm	Signatures
[Li et al., 2020]	CGAN	LSTM	KL divergence	Limit order books
PCFGAN [Ni et al 2023]	GAN	NN	Characteristic fn	Time series
ExGAN [Bhatia et al., 2020]	CGAN	CNN	KL divergence + Supervised loss	Rainfall images

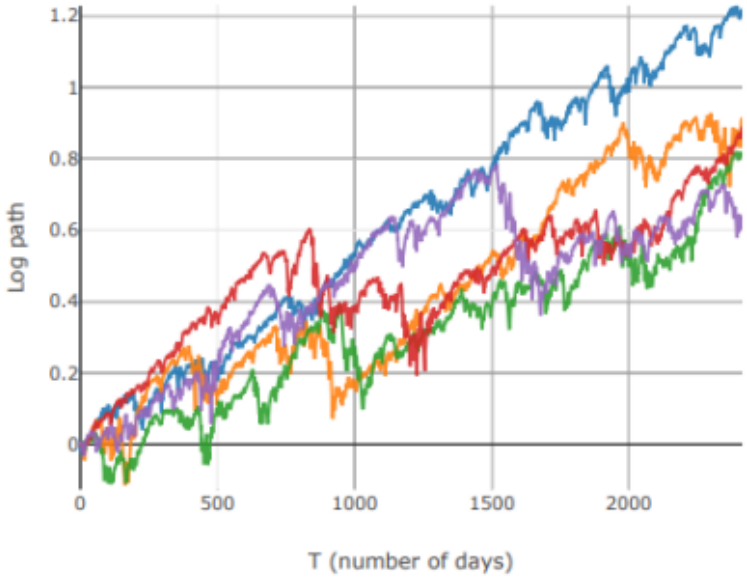
Figure: Table 1: Literature review.

Key questions in model validation

How do we know if the generated scenarios are realistic?

- In the context of images or text, user perception can be easily used for validation
- virtually check is impossible to evaluate the simulated finance scenarios

(a) Image generation of dogs



(b) Financial scenarios: which path is a possible realization of S&P500?

Validation Criterion

Criterion should be tailored to the down-stream tasks of interest.

Prior works focus on "global properties" (such as Wasserstein distance) of the return distribution:

- Global property v.s. Tail behaviour (rare events)?
- Raw return distribution v.s. PnL distribution of trading strategies?

Overview of TAIL-GAN framework

Tail risk and Elicitability:

- Value at Risk (VaR) and Expected Shortfall (ES) are commonly used statistics for measuring the tail risk of a variety of trading strategies
- VaR and ES of a group of trading strategies are incorporated in the loss function by utilizing the joint elicitability property

Discrimination via dynamic trading strategies:

- static portfolios are used to capture cross-asset dependence information
- dynamic strategies are used to capture temporal dependence information in the financial scenarios

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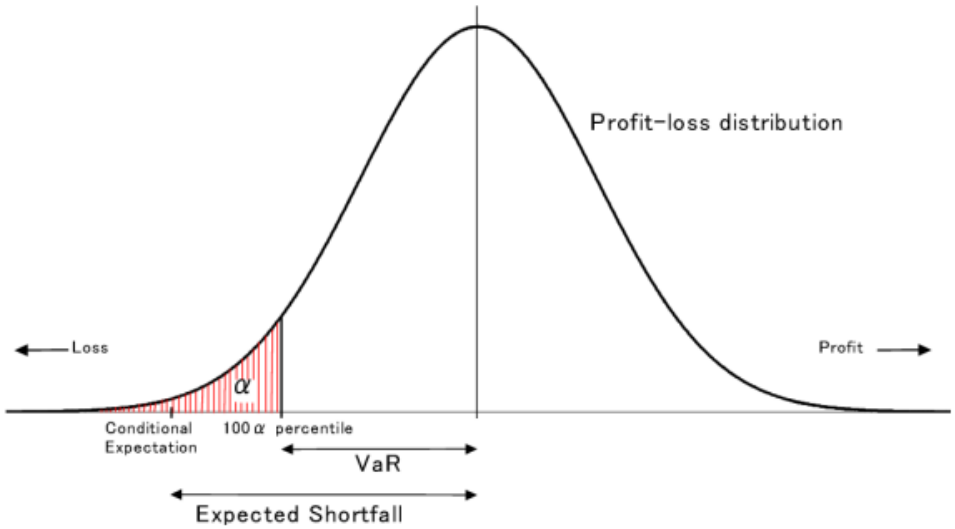
Tail risk measures

Value-at-Risk (VaR) and Expected Shortfall (ES)

Given a one-dimensional distribution μ with a finite mean and a confidence level $0 < \alpha < 1$, the α-VaR and α-ES are defined as

$$\mathrm{VaR}_\alpha(\mu) := \inf\{\, x \in \mathbb{R} : \mu(X \le x) \ge \alpha \,\}$$

$$\mathrm{ES}_\alpha(\mu) := \frac{1}{\alpha} \int_0^\alpha \mathrm{VaR}_\beta(\mu)\, \mathrm{d}\beta$$



Elicitability: from statistical estimation to optimization

Elicitability

A statistical functional T (e.g., mean) is **elicitable** for a set of distributions $\mathcal{F}$ if there is a score function $S(x, y)$ such that

$$T(\mu) = \arg\min_x \int S(x, y)\, \mu(dy)$$

for any distribution $\mu \in \mathcal{F}$.

Examples:

- mean with $S(x, y) = (x - y)^2$
- median with $S(x, y) = |x - y|$
- ES_α is not elicitable but (VaR_α, ES_α) is jointly elicitable
- Spectral risk measure $\Rightarrow$ allows us to incorporate multiple quantile levels

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Some notations

- Consider the price dynamics of M assets over a time horizon T.
- The user specifies K trading strategies which may be static or dynamic.
- $\Pi^k : \mathbb{R}^{M \times T} \rightarrow \mathbb{R}$ represents the k-th strategy, with the price scenarios $\mathbf{p} \in \mathbb{R}^{M \times T}$ as the input and the final PnL as the output.
- $\Pi^k \# \mathbb{P}_r$ denotes the distribution of the PnL of the k-th strategy $\Pi^k(\mathbf{p})$, where $\mathbf{p} \sim \mathbb{P}_r$.

Theoretical version of the discriminator

- $D : \mathcal{P}(\mathbb{R}) \to \mathbb{R}^2$ a mapping parameterized by a neural network, with strategy PnLs as the input and two values as the output
- **Goal:**

$$\overbrace{D\Bigg(\underbrace{\Pi^k \# \mathbb{P}_r}_{\text{strategy PnL distribution}}\Bigg)}^{\text{VaR and ES prediction from D}} \approx \big(\mathrm{VaR}_\alpha(\Pi^k \# \mathbb{P}_r),\, ES_\alpha(\Pi^k \# \mathbb{P}_r)\big)$$

- **Mathematically,**

$$D^* \in \arg\min_D \frac{1}{K} \sum_{k=1}^{K} \mathbb{E}_{p \sim \mathbb{P}_r}\Big[\, S_\alpha\big(D(\Pi^k \# \mathbb{P}_r); \mathbf{p}\big) \,\Big]$$

The generator

- Input: random noise $\mathbf{z} \sim \mathbb{P}_z$ distribution on the latent space
- Output $G(\mathbf{z})$: simulated price scenario
- Mapping parameterized by neural network:

$$G(\mathbf{z}; \mathbf{W}, \mathbf{b}) = \mathbf{W}_L \, \sigma\Big(\mathbf{W}_{L-1} \cdots \sigma(\mathbf{W}_1 \mathbf{z} + \mathbf{b}_1)\Big) + \mathbf{b}_L,$$

where

- $\mathbf{W} = (\mathbf{W}_1, \ldots, \mathbf{W}_L)$ and $\mathbf{b} = (\mathbf{b}_1, \ldots, \mathbf{b}_L)$ represent the trainable parameters, with $\mathbf{W}_\ell \in \mathbb{R}^{n_\ell \times n_{\ell-1}}$ and $\mathbf{b}_\ell \in \mathbb{R}^{n_\ell \times 1}$ for $\ell = 1, 2, \ldots, L$.
- The operator $\sigma(\cdot)$ is a component-wise non-linear operator.

Objective of the generator

The goal of the generator is to best match the tail risks across the set of K benchmark strategies:

$$\underbrace{D(\Pi^k \# \mathbb{P}_r)}_{\text{strategy PnL distribution}} \approx \Big(\mathrm{VaR}_\alpha(\Pi^k \# \mathbb{P}_r),\ ES_\alpha(\Pi^k \# \mathbb{P}_r)\Big)$$

Objective of the generator

The goal of the generator is to best match the tail risks across the set of K benchmark strategies:

$$\overbrace{D^*(\Pi^k \# \mathbb{P}_G) \approx D^*(\Pi^k \# \mathbb{P}_r)}^{\text{Goal of the generator}} \approx \underbrace{\Big(\mathrm{VaR}_\alpha(\Pi^k \# \mathbb{P}_r),\ \mathrm{ES}_\alpha(\Pi^k \# \mathbb{P}_r)\Big)}_{\text{Goal of the discriminator}}$$

Mathematically,

$$\min_G \frac{1}{K} \sum_{k=1}^{K} \mathbb{E}_{\mathbf{p} \sim \mathbb{P}_r} \Big[S_\alpha \Big(\underbrace{D^*(\Pi^k \# \mathbb{P}_G)}_{\text{VaR and ES prediction from } D^*} ;\ \Pi^k(\mathbf{p}) \Big) \Big]$$

Universal approximation of the generator

Theorem (Cont et al. [2025]):

Assume mild Lipschitz and regularity conditions. For any given error $\varepsilon > 0$, there exists a fully connected and feed-forward deep neural network G_1 with ReLU activation such that

$$\left|\mathrm{VaR}_\alpha\Big(\Pi^k\#((\nabla G_1)\#\mathbb{P}_z)\Big) \;-\; \mathrm{VaR}_\alpha\Big(\Pi^k\#\mathbb{P}_r\Big)\right| < \varepsilon.$$

for all $1 \le k \le K$. Here

- The depth of G_1: $L = \lceil \log n_1 \rceil$;
- The width of G_1: $N = 2^L = 2^{\lceil \log n_1 \rceil}$;
- $n_1 = \mathcal{O}(\varepsilon^{-2})$.

Universal approximation of the generator

Theorem (Cont et al. [2025])

Assume mild Lipschitz and regularity conditions ($\mathbb{P}_r$ has bounded βth moment $1 < \beta \leq 2$). For any given error $\varepsilon > 0$, there exists a fully connected and feed-forward deep neural network G_2 with ReLU activation such that

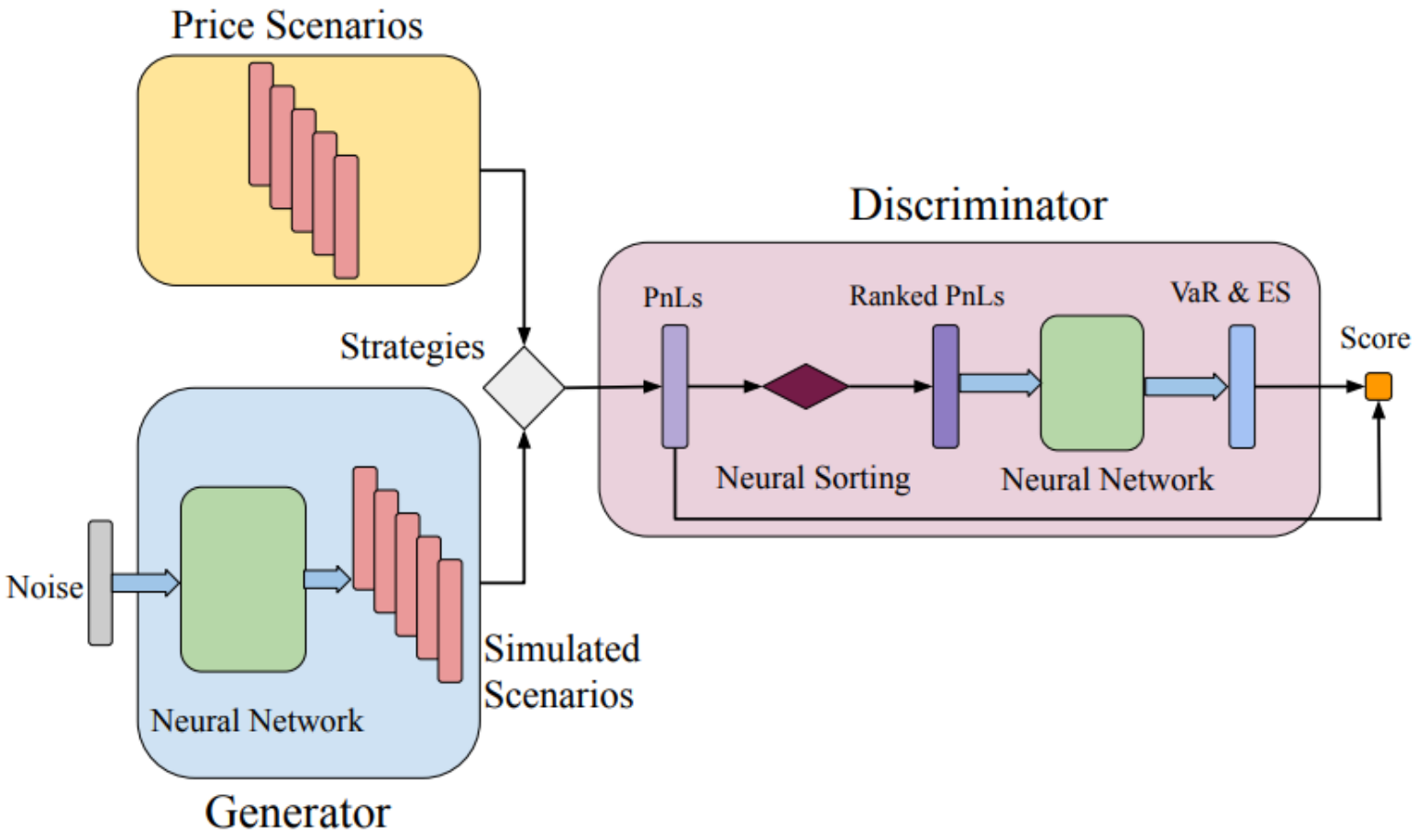
$$\left|\mathrm{ES}_\alpha\big(\Pi^k \#\big((\nabla G_2)\#\mathbb{P}_z\big)\big) - \mathrm{ES}_\alpha\big(\Pi^k \#\mathbb{P}_r\big)\right| < \varepsilon.$$

for all $1 \leq k \leq K$. Here

- *The depth of G_2: $L = \lceil \log n_2 \rceil$.*
- *The width of G_2: $N = 2^L = 2^{\lceil \log n_2 \rceil}$.*
- $n_2 = \mathcal{O}\big(\varepsilon^{-\beta/(\beta-1)}\big)$.

- Similar results also hold for risk measures that are Hölder continuous under the Wasserstein measure.

Summary of the architecture



From bi-level optimization to max–min game

Our goal is to find a generator G^* and a discriminator D^* via the following bi-level optimization problem

$$G^* \in \arg\min_G \frac{1}{K}\sum_{k=1}^{K} \mathbb{E}_{\mathbf{p}\sim\mathbb{P}_r}\Big[S_\alpha\big(D^*(\Pi^k \# \mathbb{P}_G),\ \Pi^k(\mathbf{p})\big)\Big],$$

where

$$D^* \in \arg\min_D \frac{1}{K}\sum_{k=1}^{K} \mathbb{E}_{\mathbf{p}\sim\mathbb{P}_r}\Big[S_\alpha\big(D(\Pi^k \# \mathbb{P}_r),\ \Pi^k(\mathbf{p})\big)\Big].$$

$\hookrightarrow$ constrained optimization problem is usually difficult to solve in practice!

From bi-level optimization to max–min game

Theorem (Cont et al. [2025])

Under mild conditions, the bi-level optimization problem is equivalent to the following max–min game:

$$\max_{D} \min_{G} \frac{1}{K} \sum_{k=1}^{K} \Bigg[\mathbb{E}_{\mathbf{p}\sim\mathbb{P}_r} \Big[S_\alpha\big(D(\Pi^k \# \mathbb{P}_G),\, \Pi^k(\mathbf{p})\big) \Big] - \lambda\, \mathbb{E}_{\mathbf{p}\sim\mathbb{P}_r} \Big[S_\alpha\big(D(\Pi^k \# \mathbb{P}_r),\, \Pi^k(\mathbf{p})\big) \Big] \Bigg].$$

with any Lagrange multiplier $\lambda > 0$.

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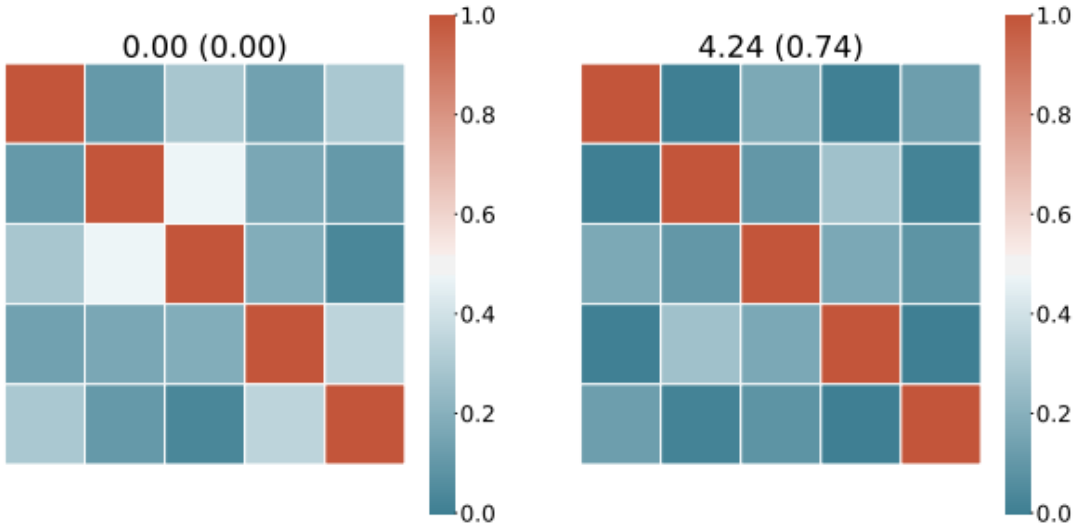
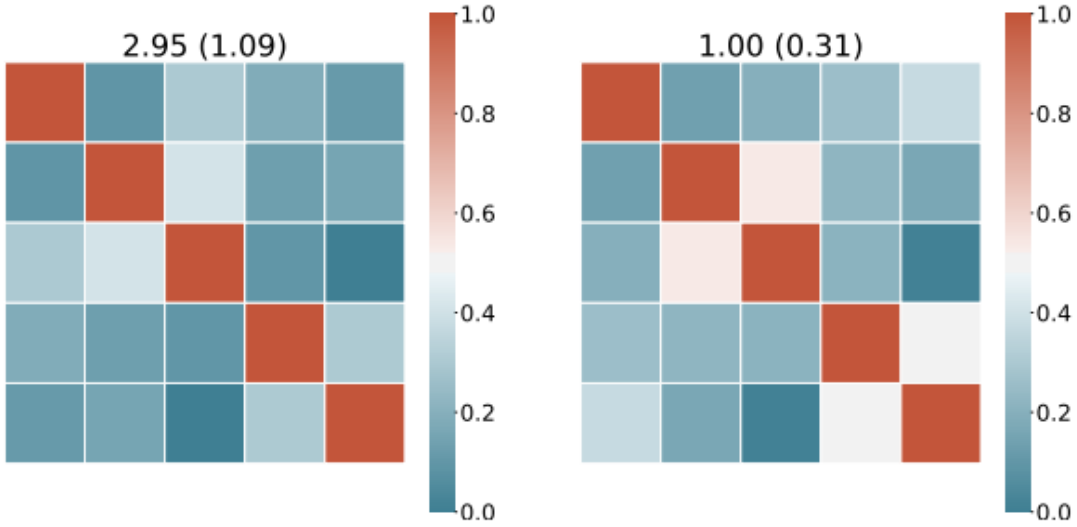
Correlation patterns

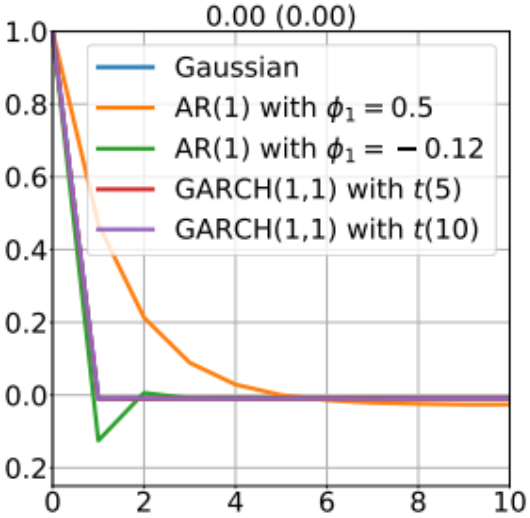


(c) Market data. (d) TAIL-GAN-Raw.

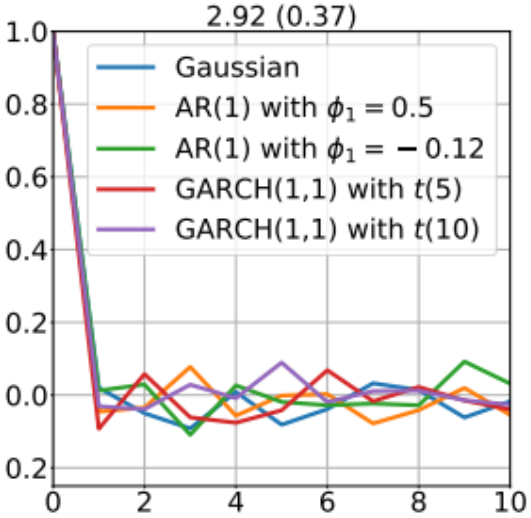


(e) TAIL-GAN-Static. (f) TAIL-GAN.

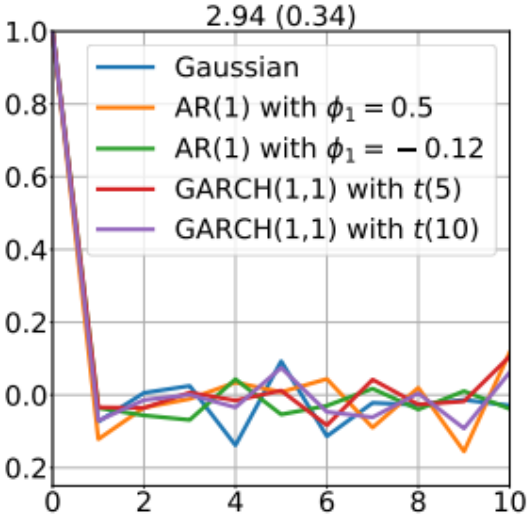
Temporal patterns



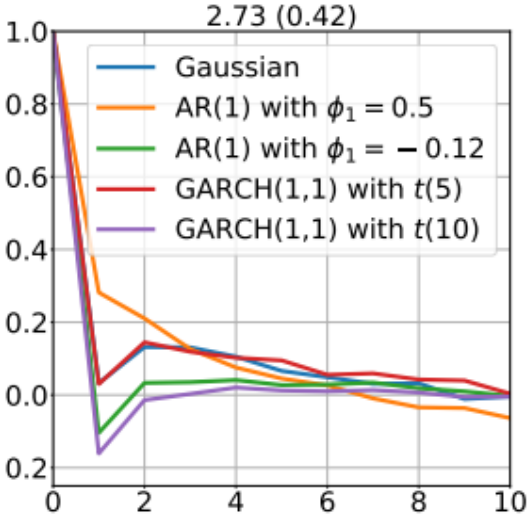
(a) Market data.



(b) TAIL-GAN-Raw.



(c) TAIL-GAN-Static.



(d) TAIL-GAN.

Performance: out-of-sample error

	"Oracle"	TAIL-GAN-Raw	TAIL-GAN-Static
%	3.0	83.3	86.7
(std)	(2.2)	(3.0)	(2.5)
	TAIL-GAN	WGAN	
	4.6	21.3	
	(1.6)	(2.2)	

Thank you!

https://pubsonline.informs.org/doi/abs/10.1287/mnsc.2023.00

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Which score function to use?

In general, the score function is not unique...

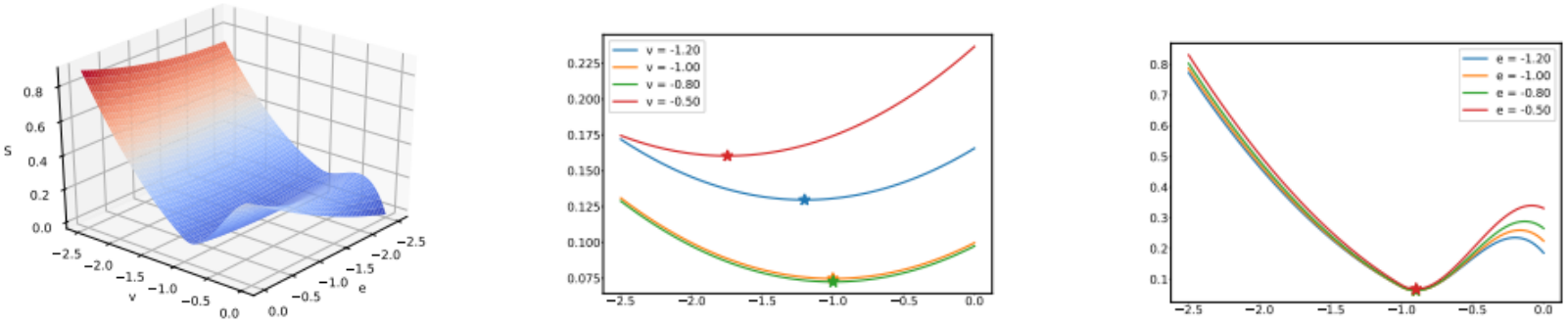
$$\begin{aligned} S_\alpha(v,e,x) &= (\mathbf{1}_{\{x\le v\}} - \alpha)\,(H_1(v) - H_1(x)) \\ &\quad + \frac{1}{\alpha} H_2'(e)\,\mathbf{1}_{\{x\le v\}}(v-x) + H_2'(e)(e-v) - H_2(e) \end{aligned}$$

- Fissler and Ziegel [2016] proposes: $H_2 : \mathbb{R} \to \mathbb{R}$ strictly convex and $R_\alpha(v,e) := \frac{1}{\alpha} v H_2'(e) + H_1(v)$ strictly increasing in v.
- Acerbi and Szekely [2014] propose: $H_1(v) = -\frac{W_\alpha}{2} v^2, \quad H_2(e) = \frac{\alpha}{2} e^2, \quad$ with $\frac{ES_\alpha(\mu)}{VaR_\alpha(\mu)} \;\ge\; W_\alpha \;\ge\; 1$

Which score function to use?

Proposition [Cont et al., 2025]

The score function proposed in Acerbi and Szekely [2014] has a better optimization landscape near the global minimum.



(a) $s_\alpha(v, e)$ (b) $s_\alpha(v, e)$ as a function of e (c) $s_\alpha(v, e)$ as a function of v

Figure 3: Landscape of $s_\alpha(v, e) = \mathbb{E}_{X \sim U([-1,1])}[S_\alpha(v, e, X)]$ based on Acerbi and Szekely [2014]